

- 2) Write down a short note on Environmental Kuznets Curve hypothesis.

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- 3) Write down a short note on relationship between Consumption and production externalities with international trade.

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17.3 TRADE AND CLIMATE CHANGE

Since nineties, the globalizing forces have significantly enhanced trade flows. The growing trade has often been linked to environmental degradation (Copeland and Taylor 2004) and climate change concerns (WTO-UNEP 2009). The influence of trade flows on climate change can be explained with help of three effects. First, rise in production and exports may necessitate greater use of energy products leading to higher emissions of GHGs, which result in *scale effect*. Secondly, if a country specializes in manufacturing sectors in general and polluting sectors in particular, adverse environmental consequences may follow, leading to *composition effect*. Finally, if economic growth is accompanied by stricter emission abatement standards and stronger regulatory policies to contain the adverse climate change concerns, a more sustainable scenario might emerge through *technique effect*.

While literature on trade flows-climate change interrelationship is quite vast, the empirical evidence does not offer a clear conclusion. Cross-country studies indicate that increased outward orientation generally leads to higher GHGs emissions and environmental degradation. In addition to merchandise trade, trade in services can also significantly influence CO₂ emissions. Several country-specific studies have also analyzed the relationship between trade and climate change through time series models or case study-based approach. Evidence on emission of methane and other GHGs from agricultural and livestock exports have been noted both in developed and developing countries. Positive relationship between growing merchandise exports and increased CO₂ emissions in several developing countries has also been reported. The possible trade-emission correlation receives further support from the *Pollution Haven Hypothesis* (PHH) literature, which argues that FDI inflows in a developing country may target the pollution-intensive sectors, thereby strengthening both the *scale* and *composition effects*.

Another section of the literature however disagrees with these results. For instance, while intra-European trade flows results innovation and technology

spillovers, their trade with developing countries might lead to GHGs emissions. The PHH has been rejected by several studies arguing that it is in the best interest of the foreign investor to bring 'greener' technology, rather than extracting the advantage of lax environmental standards. Awareness is rising in many developing countries in recent period for reducing carbon emissions from trade-related activities through ISO 14000 certifications, technology upgradation etc. In all, while growing importance of *technique effect* in OECD and other high-income countries has been noted, the same might be relatively weaker for their lower income counterparts.

17.4 TRADE AND ENVIRONMENT: POLICY AND PRACTICE

The findings of the existing literature indicates that the choice of environmental policy instruments for international trade is often constrained by identification of stage of pollution generation, i.e., whether the pollution is production-based and / or consumption-induced in nature. In addition, environmental externalities of international trade could also turn out to be both production and consumption related. Trade induced *composition effects* influence pollution from production and consumption differently. Moreover, sourcing inputs for production often causes environmental problems, which are mostly local. In practice, many of the environmental impacts of international trade are local in nature (e.g., water and air pollution, land degradation, deforestation) and often exporting firms ignore local environmental problems to achieve export competitiveness. This in effect enables them to have a lower compliance cost for doing business. However, this perspective offers a moral argument to the importing countries for laying stringent certification criteria for their imports. It is argued that right choice of environmental policy instruments depends on the characteristic of the imported goods, in particular, whether they are substitutes (perfect / imperfect) or complements as well as the underlying international market condition for that product, i.e., whether perfect competition or monopoly prevails.

The implications of using environmental tax as a policy instrument have been a widely researched area. However, the advantages of linking such measures with trade have been debated. If environmental policy instruments are introduced to address the causes of environmental degradation based on the principles of environmental law (e.g., 'rectification at source', 'precautionary action', 'prevention action', 'polluter-pays'), the intervention should occur either at the production level and/ or at the consumption level rather than at the trade level. While imposition of environmental taxes improves welfare, direct setting of standards under certain circumstances may lead to welfare loss. Given the institutional capacity, input taxes on polluting inputs could achieve substantial abatement while minimizing monitoring costs. These instruments are, 'much more targeted in the sense that they are likely to achieve substantial abatement of pollution while inducing smaller welfare losses, because they have a less-

distorting impact on the rest of the economy'. The roles of structured fiscal policies (emissions taxes or their cap-and-trade equivalents with allowance auctions) for promoting greener economies are desirable. There are recommendations to impose tax on coal, oil products, and natural gas, given the potential CO₂ and local emissions from these sectors.

Import tariff reduction makes inputs cheaper, which in turn promotes production. As such a policy increases the marginal social cost; there is an incentive to impose stricter environmental regulations. Conversely, as tariff reduction lowers the revenue collection, a case for maintaining a set of weaker regulations arises that will lower the compliance cost for domestic players. While the relative strengths of the two opposing forces are ambiguous, it is expected that import tariff liberalization may not necessarily degrade the environmental scenario.

One major constraint to implement environmental policy instruments with trade objective is that they may lead to capital flight and relocation of firms, with growth implications for the host country. The perceived consequences might be more severe in case of developing countries and LDCs in general and the Asian context in particular. Evidence shows that FDI inflows have often targeted polluting industries in line with the PHH phenomenon in several South (e.g., India), East (e.g., China) and Southeast (e.g., Malaysia, Thailand, the Philippines) Asian countries. In addition, developing economies have limited institutional capacity to implement environmental policies which often constrains their ability to choose from available policy instruments for environmental protection. In face of institutional limitations, developing economies often choose tariff / trade barrier to protect domestic environment, though it cannot target pollution adequately and it has opposite effects on consumption-induced and production-based pollution.

As the converse of the tax policies, there exist a rich literature on provisions of subsidies and their implications on exports. Provision of subsidies *per se* does not imply a challenge to environmental sustainability. However, countries often provide subsidies in the area of agricultural products and natural resources, as well as offering energy subsidies to manufacturing products. Both the routes may exert an adverse impact on local sustainability as well as global climate change concerns. For instance, subsidy-led adverse environmental impacts are often noted in resource-intensive sectors like primary activities (e.g., agriculture, fisheries), transport, energy and water, etc. Also, greater fiscal devolution is often associated with rise in increasing forest land clearing for agricultural production. Fuel subsidies received by fishing trawlers and other vessels often lead to overuse of mineral fuels and over-fishing. , The provision of fuel subsidies to the iron and steel industries in China, is another case in point. On the whole, it is observed that higher budgetary spending generally leads to rise in CO₂ emissions.

In the existing literature there is no common consensus that international trade is detrimental for environment and the results are sensitive to different countries

and pollutants. For instance, there are many studies which support PHH but many studies refute it too. Often the findings are contingent on choice of methodology for data analysis. Larger data availability for a broad set of pollutants for many years and countries could facilitate generalization of the findings. In presence of these constraints, adopting proactive actions to mitigate global environmental problems like GHGs emissions often lead to adoption of environmental policy instruments to negate the environmental impacts of international trade. Underlying economic rationality behind adoption of these environmental policy instruments is that they could take into account the differences between social and private marginal costs of production relating to international trade-related activities and at the best could capture the marginal cost of abatement of pollution. However, the design of environmental policy instrument plays an important role, given its potential to distort the market conditions and possible emergence as trade barriers.

The compliance cost related concerns on competitiveness may overshadow the actions of key players, thereby affecting the process of both trade as well as climate change negotiations.

The perceived need for imposing border trade adjustments (BTAs) on imports for tackling global climate change concerns are increasingly being explored by the countries in this context and a rich literature on this front has come up. BTAs can be defined as, 'requiring trade taxes to partially reflect emission leakage effect arising from asymmetric environmental policies or, more narrowly, as requiring trade taxes to specifically embody a nation's carbon tax on the carbon content of imports'. BTAs are instruments to provide a level playing field to domestic producers where environmental standards are stringent. In practice, BTAs are charged on imports equivalent to what domestic producers pay to protect the environment and exporters are rebated for the cost they incur.

The impacts of BTAs are ambiguous as on one hand, they can potentially solve the problem of carbon leakage, encourage developing countries to protect global environment, help to achieve economic efficiency in carbon abatement, improve global welfare as it transfers carbon pricing, address the competitiveness concerns of developed countries, but also may inflict damage upon the trade of developing countries on the other. Many scholars acknowledge the importance of intervention on the part of the importing country given the fact that carbon prices differ internationally and carbon taxes in some countries might be constrained. It further argued adding to the policy debate that the, 'efficiency case for BTA depends critically on whether climate policies are pursued by carbon taxation or by cap-and-trade'. Scholars also support the imposition of BTAs, given the potential incentive for countries to weaken their standards to enhance their market access.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Explain the role of an effective environment policy in reducing environmental hazards.

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- 2) Explain the major constraint to implement environmental policy instruments.

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17.5 ROLE OF WTO TO SAFEGUARD ENVIRONMENT

The role of WTO in achieving sustainable development and protecting environment is explained in the following WTO statement:

“Allowing for the optimal use of the world’s resources in accordance with the objective of sustainable development and seeking to protect and preserve the environment are fundamental to the WTO. These goals, enshrined in the Preamble of the Marrakesh Agreement, go hand in hand with the WTO’s objective to reduce trade barriers and eliminate discriminatory treatment in international trade relations. For WTO members, the aims of upholding and safeguarding an open and non-discriminatory multilateral trading system, on the one hand, and acting for the protection of the environment and the promotion of sustainable development, on the other, can and must be mutually supportive.”³

Ensuring that trade contributes to sustainable development is an explicit objective of the WTO and the 2030 Agenda for Sustainable Development (including the Sustainable Development Goals or ‘SDGs’). However, one area of concern is that the legal provisions of the multilateral trade regime, i.e., World Trade Organization (WTO), were not originally created to deliberate on climate change policies and sustainability challenges. While on one hand the text of the WTO Sanitary and Phytosanitary (SPS) and Technical Barriers to Trade (TBT) agreements advocate against the use of unilateral trade policies and base the

³ An introduction to trade and environment in the WTO, available online at: https://www.wto.org/english/tratop_e/envir_e/envt_intro_e.htm (last accessed on 31 July 2017).

actions on 'scientific evidence', the 'precautionary principles' based on the occurrence of 'irreversible damage' are also enshrined among the provisions.⁴ The non-recognition of 'polluter-pays-principle' by international trade institutions pose a challenge for imposing unilateral policies addressing cross-border sustainability challenges. The analysis further noted that as such bodies do not recognize the economic equivalence of emission tax and permit regimes, the resulting complexities make certain environmental policies susceptible to influence by protectionist motives. Hence the WTO-compatibility of BTAs is still open to debate.

So far, the analysis presents a case for importing countries to impose stricter environmental standards to support the climate change concerns and to protect the competitive concerns of the domestic players on the other. Nevertheless, the possibility of such measures functioning as *de facto* non-tariff barriers (NTBs) on exports of developing countries has been raised at times. In order to comply with strict environmental standards in the 'North', the producers in 'South' need to undertake compliance costs on multiple fronts, which may lead to loss of their competitiveness. If a developing country is forced to adopt a stricter environmental standard set by an external entity for exporting, its export sector suffers adversely. The underlying motive behind imposition of such standards has also been questioned. Higher standards imposed by developed countries always increase their domestic firms' profitability, but reduce the same for firms located in developing countries. In addition, it has been argued that unilateral environmental actions / sanctions are generally ineffective as an enforcement device.

The other branch of literature has noted that the stringent environmental standards have significantly improved the sustainability scenario in the South. For instance, ISO 14001 certification on environmental management awards has witnessed a 1,500 per cent increase between 1999 and 2009. The adverse effects on exports have also been questioned. It has been shown in the Chinese context that barring the exception of chemical industry, stricter environmental regulation has not adversely influenced the export potential. On the contrary, the quality of exports from both resource-intensive primary and high-value-added manufacturing sectors has improved.

In the past, trade and environment related disputes have been raised at the WTO dispute settlement forums and a 'North' move towards BTAs would only increase the number further. Given the absence of a clear understandings and agreement at the global level on managing the climate change effects through trade instruments, it is clear that the discord over environmental issues may lead

⁴ In principle, sanitary and phytosanitary (SPS) measures aim to protect the health of humans, plants and animals, while technical barriers to trade (TBT) ensure product quality and safety. However, governments may overshoot the requirements of health and consumer safety and use SPS and TBT to shield domestic producers from fair competition.

to welfare loss. The WTO case law has provided a mixed view so far. For instance, certain aspects of the verdict in DS 58⁵ pave the way for increasing incidence of unilateral trade sanctions in the future. On the other hand, by disapproving the unilateral action of the EU in the biotech case (DS 291-293)⁶, a strong message against adoption of such measures has been sent. Taking note of the legal issues yet to be resolved under WTO, the need to enable and encourage good policymaking is warranted.

The potential for a country to use voluntary export restraints (VERs) on its exports with sustainability undercurrent has recently been observed at WTO in the *rare earth* dispute. China's decision to limit export of rare earth materials, which are crucial sub-components for electronics and other high-tech products, citing their overexploitation was met by protests from countries where the domestic industries were crucially dependent on Chinese imports. Disputes were lodged at the WTO against the Chinese adoption of various policy instruments to further this goal: export duties, export quotas, export licensing and minimum export price requirements. WTO Dispute Settlement Panel ruled against China by noting that some of their instruments were geared to achieve industrial policy objectives rather than sustainability.⁷ The precedence of this case will have profound implications for other trade-environment disputes lodged at the WTO in days to come.

Protection of environment and promotion of sustainable development (SD) is one of the mandates of WTO (WTO undated). The environmental provisions are also enshrined under several WTO agreements. A change in the existing provisions is however the need of the hour. There are four specific areas for this purpose – allowing production subsidies for certain green products and technologies; arriving at a possible compromise on BTA based on carbon content in domestic production, removing / relaxing existing restrictions on exports of natural gas to facilitate substitution of dirtier substitutes like oil and coal, and tightening compulsory license mechanism for green technologies in emerging economies. The third point becomes particularly important, as a number of Asian countries

⁵ See, Dispute Settlement Case No. 58, *United States — Import Prohibition of Certain Shrimp and Shrimp Products (complaint by India; Malaysia; Pakistan; Thailand)* for details.

⁶ See, Dispute Settlement Case No. 291-293, *European Communities — Measures Affecting the Approval and Marketing of Biotech Products (complaint by United States; Canada; Argentina)* for details.

⁷ For details see, Dispute Settlement Case No. 431, *China — Measures related to the Exportation of Rare Earths, Tungsten and Molybdenum* (Complainant: United States); Dispute Settlement Case No. 432, *China — Measures related to the Exportation of Rare Earths, Tungsten and Molybdenum* (Complainant: European Union); Dispute Settlement Case No. 433, *China — Measures related to the Exportation of Rare Earths, Tungsten and Molybdenum* (Complainant: Japan).

have recently been at the WTO dispute settlement forums on the ground of support to the renewable energy sector.⁸

17.6 MULTILATERAL ENVIRONMENT AGREEMENTS AND TRADE

The role of international cooperation (agreement) is inevitable for many environmental problems. The first international treaty on trade and environment was the Phylloxera Agreement of 1978 to restrict trade in grapevines to prevent the spread of pests that damage vineyards. In response to emerging environmental challenges, a large number of multilateral environmental agreements (MEAs) were mooted to minimize the trade related impacts on environment and natural resources. These agreements played important role in conservation of natural resources and biodiversity, protecting endangered species, and minimizing spillover of hazardous substances and environmental problems across countries. About 17 MEAs were identified by GATT (1992) to contain trade measures, starting in 1933. However, each of these have not ratified by all the countries. Major MEAs having bearing on trade and environment are as follows:

- International Convention for the Protection of Birds (1950)
- International Plant Protection Agreement (1951)
- Convention on Conservation of North Pacific Fur Seals (1957)
- Agreement Concerning Cooperation in the Quarantine of Plants and their Protection against Pests and Diseases (1959)
- Rio International Convention for the Conservation of Atlantic Tunas (ICCAT) (1966)
- Phyto-sanitary Convention for Africa (1967)
- The African Convention on the Conservation of Nature and Natural Resources (1968)
- The Benelux Convention of on the Hunting and Protection of Birds (1970)
- Convention on International Trade in Endangered Species (CITES) of Wild Fauna and Flora (1973)
- Montreal Protocol on Substances that Deplete the Ozone Layer (1987)

⁸ For details see, Dispute Settlement Case No. 412, *Canada — Certain Measures Affecting the Renewable Energy Generation Sector (complaint by Japan)*; Dispute Settlement Case No. 419, *China — Measures concerning wind power equipment (complaint by United States)*; Dispute Settlement Case No. 456, *India — Certain Measures Relating to Solar Cells and Solar Modules (Complainant: United States)*.

- Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (1989)
- Convention for the Prohibition of Fishing with Long Driftness in the South Pacific (Wellington Convention) (1990)
- Antarctica Treaty (1991)
- Convention on Biological Diversity (1992)
- Convention on Straddling and Highly Migratory Fish Stocks (1995)
- Kyoto Protocol on Climate Change (1997)
- Stockholm Convention on Persistent Organic Pollutants (2001)
- Convention on Biological Diversity (2002)
- Cartagena Protocol on Biosafety (2003)

Since all MEAs are not compatible with WTO rules, the precedence of international agreements in case of conflict is a matter of debate.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Comment on the role of WTO in safeguarding environment..

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2) Explain the major multilateral agreements to control environmental degradation.

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17.7 LET US SUM UP

In this unit, we discussed about linkages between trade and environment. International trade itself may not automatically correct the distortions in prices, thereby making environmental policy instruments one of the options available. Achieving high economic growth and exports, creating jobs, etc. are often the foremost priorities compared to protecting the environment for many developing and emerging economies in the world, and the Asian economies are no exception to this trend. Therefore, there is lack of seriousness to implement commitments made in multilateral environmental agreements. Though regional forums exist to discuss sustainability concerns, the working of the same in the South is still in the

evolving stage. While environmental sustainability has been included among the areas of cooperation in the Asian regional trade agreements (RTAs), they are still not appropriate forums to hold a country for particular domestic environmental concerns. The contrast with inclusion of environmental provisions under North American Free Trade Agreement (NAFTA) or the EU Economic Partnership Agreements (EPAs) is clearly observed. This is one area for the Asian countries to deepen regional cooperation. Regional Comprehensive Economic Partnership (RCEP) may play a key role in this regard.

As several Asian countries are now facing environmental compliance requirements in developed countries, which often function as NTBs their exports, there is a visible move towards carbon tax regimes - e.g., pilot emissions trading schemes (ETS) at municipal and provincial levels in China, introduction of ETS in South Korea, introduction of the tax for climate change mitigation and voluntary emissions trading scheme in Japan, implementation of clean development mechanism (CDM) projects and a domestic voluntary carbon market in Vietnam, introduction of coal cess and pilot ETS for air pollutants in India, the proposed ETS in Indonesia and Thailand. It is expected that in the coming days, more Asian countries would be interested to tread along the path of environmental policy instruments for two reasons. First, as the United Nations Framework Convention on Climate Change (UNFCCC) negotiation intensifies, the Asian countries would also be compelled to fulfill their commitments. Second, the need to secure the export market may cause them to tighten their environmental norms and policy instruments, increasingly aligning the same with the requirements prevailing in their developed counterparts.

17.8 KEY WORDS

Environmental hazards

An environmental hazard is a substance, state or event which has the potential to threaten the surrounding natural environment / or adversely affect people's health, including pollution and natural disasters such as storms and earthquakes.

Greenhouse gases

A greenhouse gas is a gas in an atmosphere that absorbs and emits radiant energy within the thermal infrared range. This process is the fundamental cause of the greenhouse effect.

Environmental Kuznets Curve hypothesis

The Environmental Kuznets Curve (EKC) hypothesis postulates an inverted-U-shaped relationship between different pollutants and per capita income, i.e., environmental pressure increases up to a certain level as income goes up; after that, it decreases